

PRÁCTICA PREDICTIVA MINERIA DE BASES DE DATOS

**CRISTIAN CAMILO BRAN ARRIAGA
SANTIAGO CARDONA VILLADA
RICHARD JAVIER MORENO MOSQUERA**

**ESPECIALIZACIÓN DE BIG DATA E INTELIGENCIA DE
NEGOCIOS**

UNIVERSIDAD CATÓLICA LUISAMIGÓ

MEDELLIN

2025

Objetivo de la minería

Predecir el tipo de fármaco (drug) que se debe administrar a un paciente afectado de rinitis alérgica según distintos parámetros/variables. Se han recogido los datos del medicamento idóneo para muchos pacientes en cuatro hospitales. Se pretende, para nuevos pacientes, determinar el mejor medicamento a probar.

Descripción de los datos

Las variables disponibles en los historiales clínicos de cada paciente son:

- **Age:** Edad
- **Sex:** Sexo
- **BP** (Blood Pressure): Tension Sanguínea.
- **Cholesterol:** nivel de colesterol.
- **Na:** Nivel de sodio en la sangre.
- **K:** Nivel de potasio en la sangre.
- **Drug:** Hay cinco fármacos posibles {DrugA, DrugB, DrugC, DrugX, DrugY}

Modelamiento

Se requiere:

1. Preparar los datos para el análisis
2. Aplicar todos los métodos vistos en clase y analizar cada resultado obtenido por WEKA. Hacer un informe indicando los parámetros óptimos utilizados en cada método y los resultados obtenidos.
3. Establecer un criterio para seleccionar el método que mejor se ajuste a las necesidades del problema. Justificar la elección.
4. Utilice la pestaña paciente_nuevos para realizar la validación del modelo.

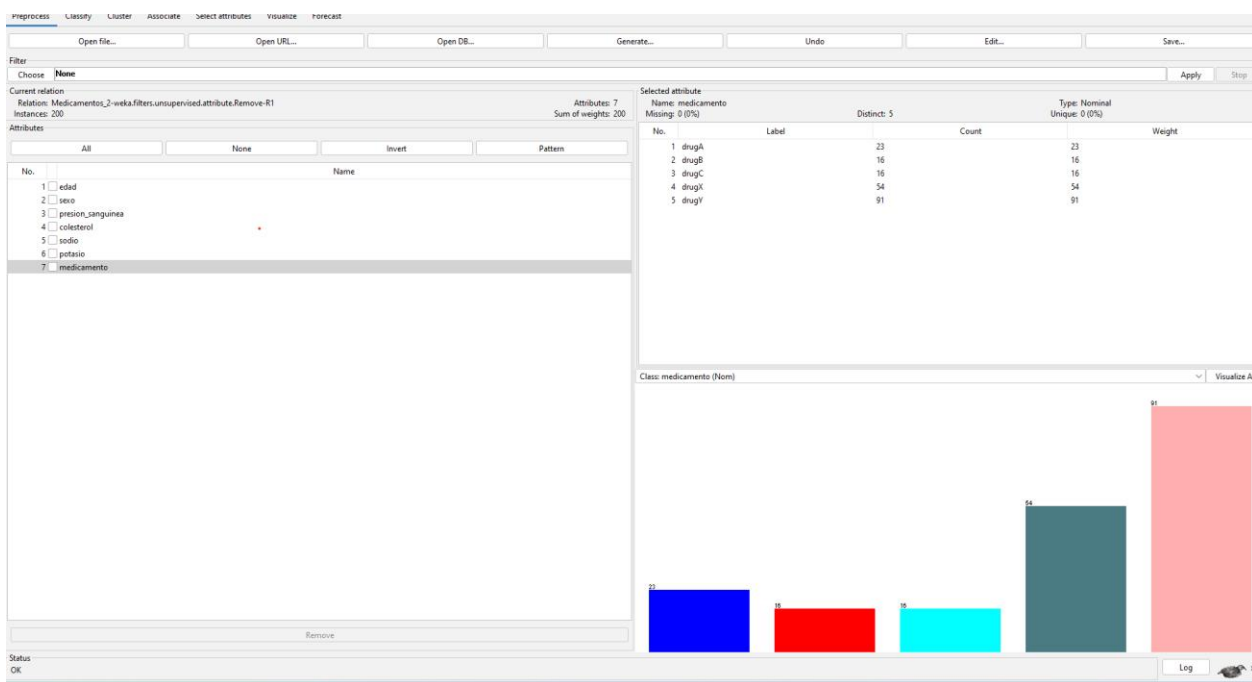
[Excel Drug](#)

[Bloc de notas Medicamentos 2](#)

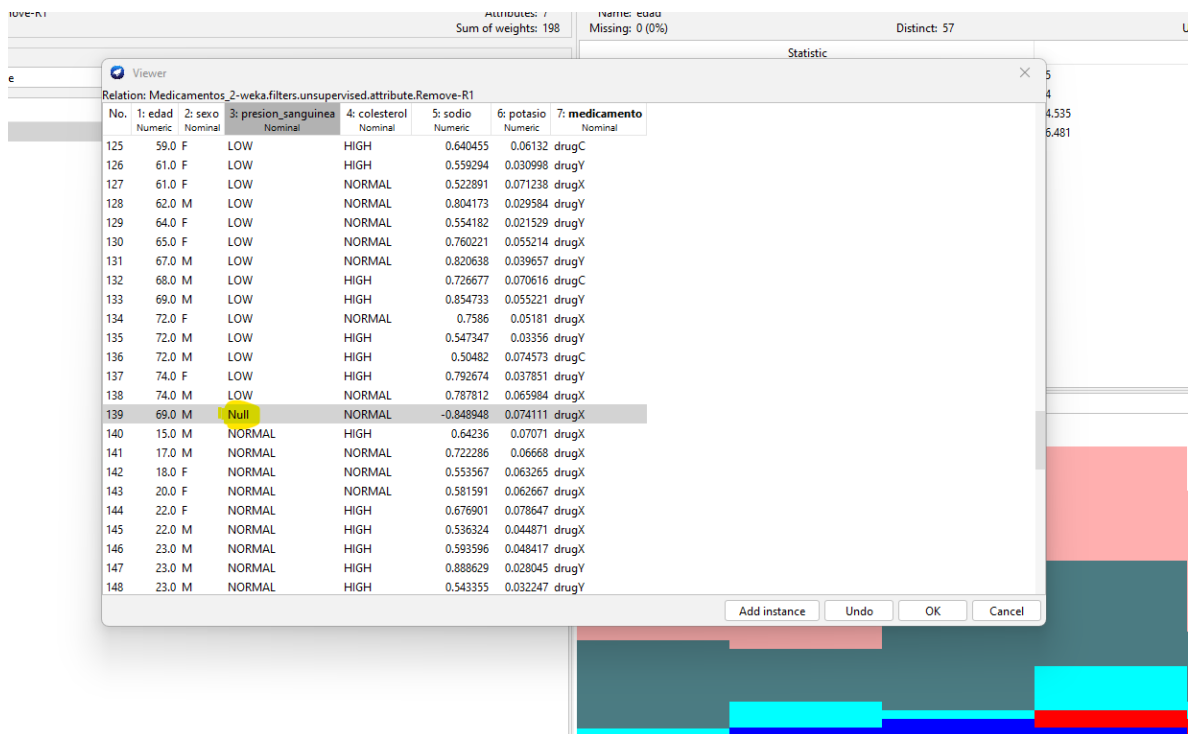
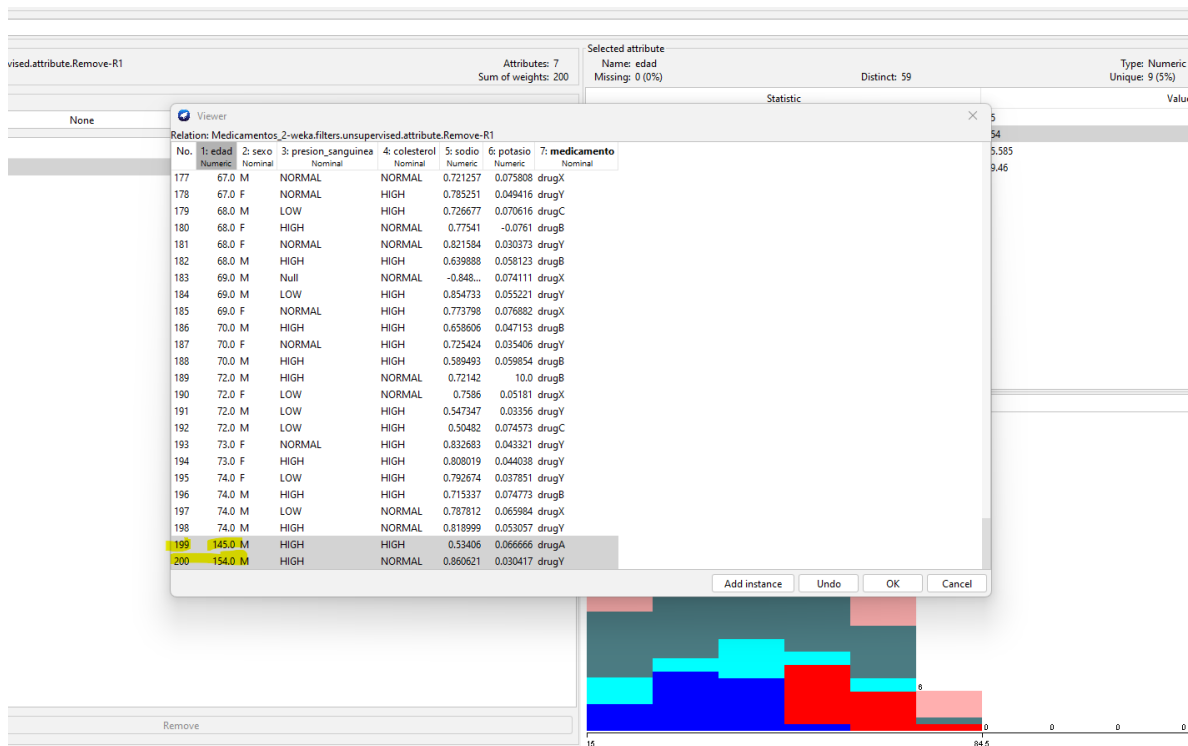
[Bloc de notas Pacientes pruebas](#)

Resultados:

1. Se inicia eliminando la variable “id” por ser un dato irrelevante.



Eliminación de datos atípicos en “edad” y “presión sanguínea”.



Se realiza el cambio de dato “Mujer” a “M”.

Sum of weights: 198 Missing: 0 (0%) Distinct: 57

Statistic

Relation: Medicamentos_2-weka.filters.unsupervised.attribute.Remove-R1

No.	1: edad	2: sexo	3: presion_sanguinea	4: colesterol	5: sodio	6: potasio	7: medicamento
	Numeric	Nominal	Nominal	Nominal	Numeric	Numeric	Nominal
76	16.0	M	LOW	HIGH	0.743021	0.061886	drugC
77	20.0	F	LOW	NORMAL	0.811023	0.069402	drugX
78	22.0	M	LOW	HIGH	0.526672	0.064617	drugC
79	23.0	M	LOW	HIGH	0.55906	0.076609	drugC
80	26.0	F	LOW	HIGH	0.578002	0.040819	drugC
81	26.0	M	LOW	NORMAL	0.790664	0.037815	drugY
82	28.0	Mujer	LOW	HIGH	0.606933	0.030659	drugY
83	28.0	F	LOW	HIGH	0.656292	0.049997	drugC
84	32.0	F	LOW	NORMAL	0.724422	0.066829	drugX
85	32.0	F	LOW	HIGH	0.730854	0.075256	drugC
86	33.0	F	LOW	HIGH	0.858387	0.025634	drugY
87	34.0	F	LOW	NORMAL	0.825542	0.063881	drugX
88	35.0	M	LOW	NORMAL	0.685143	0.074717	drugX
89	36.0	M	LOW	NORMAL	0.52765	0.046188	drugX
90	37.0	M	LOW	NORMAL	0.616692	0.068765	drugX
91	37.0	F	LOW	NORMAL	0.804155	0.066981	drugX
92	37.0	M	LOW	NORMAL	0.73154	0.043743	drugY
93	38.0	F	LOW	NORMAL	0.598753	0.020042	drugY
94	38.0	M	LOW	HIGH	0.851019	0.046516	drugY
95	39.0	F	LOW	NORMAL	0.649096	0.028598	drugY
96	39.0	M	LOW	NORMAL	0.604973	0.043404	drugX
97	40.0	F	LOW	NORMAL	0.683503	0.060226	drugX
98	41.0	M	LOW	HIGH	0.766635	0.069461	drugC
99	41.0	F	LOW	NORMAL	0.749905	0.040018	drugY

Add instance Undo OK Cancel

Se cambian los datos de “Potasio” que se encontraban en estado negativo.

upervised.attribute.Remove-R1 Attributes: 7 Sum of weights: 197 Selected attribute Name: potasio Missing: 0 (0%) Distinct: 197

None

Relation: Medicamentos_2-weka.filters.unsupervised.attribute.Remove-R1

No.	1: edad	2: sexo	3: presion_sanguinea	4: colesterol	5: sodio	6: potasio	7: medicamento
	Numeric	Nominal	Nominal	Nominal	Numeric	Numeric	Nominal
1	68.0	F	HIGH	NORMAL	0.77541	-0.0761	drugB
2	58.0	F	LOW	HIGH	0.886865	-0.023188	drugY
3	40.0	M	HIGH	HIGH	0.557133	0.020022	drugY
4	38.0	F	LOW	NORMAL	0.598753	0.020042	drugY
5	52.0	M	LOW	NORMAL	0.663146	0.020143	drugY
6	54.0	M	NORMAL	HIGH	0.504995	0.02048	drugY
7	57.0	F	NORMAL	NORMAL	0.551967	0.021317	drugY
8	20.0	M	HIGH	NORMAL	0.764067	0.021439	drugY
9	64.0	F	LOW	NORMAL	0.554182	0.021529	drugY
10	28.0	M	NORMAL	HIGH	0.584179	0.021585	drugY
11	46.0	F	HIGH	HIGH	0.773569	0.022302	drugY
12	18.0	F	HIGH	NORMAL	0.564811	0.023266	drugY
13	63.0	M	NORMAL	HIGH	0.616117	0.023773	drugY
14	18.0	F	HIGH	HIGH	0.88515	0.023802	drugY
15	31.0	M	HIGH	HIGH	0.740936	0.0244	drugY
16	65.0	M	HIGH	NORMAL	0.8645	0.024702	drugY
17	32.0	F	HIGH	NORMAL	0.643455	0.024773	drugY
18	61.0	F	HIGH	HIGH	0.63126	0.02478	drugY
19	47.0	M	LOW	NORMAL	0.84773	0.025274	drugY
20	42.0	F	HIGH	HIGH	0.533228	0.025348	drugY
21	33.0	F	LOW	HIGH	0.858387	0.025634	drugY
22	65.0	F	HIGH	NORMAL	0.828898	0.026004	drugY
23	21.0	F	HIGH	NORMAL	0.745098	0.026023	drugY
24	42.0	F	LOW	NORMAL	0.763404	0.026081	drugY

Add instance Undo OK Cancel

[illegible]

Luego se realiza normalización de datos en rangos de 0-1.

The screenshot shows the WEKA 'Attribute' window. On the left, a list of attributes includes 'edad', 'sexo', 'presion_sanguinea', 'colesterol', 'sodio', 'potasio', and 'medicamento'. The 'estad' attribute is selected. On the right, the 'Statistics' tab is active, showing the following values:

Statistic	Value
Minimum	0
Maximum	1
Mean	0.521
StdDev	0.261

Balanceo hacia arriba.

The screenshot shows the WEKA 'Selected attribute' window for the 'medicamento' attribute. It displays a table with 5 distinct values and their counts, and a bar chart visualization below it.

No.	Label	Count	Type: Nominal	Weight
1	drugA	90	Unique: 0 (0%)	90
2	drugB	90		90
3	drugC	90		90
4	drugI	90		90
5	drugY	90		90

Below the table, a bar chart titled 'Class: medicamento (Nom)' shows five bars of equal height (90) for each class, colored blue, red, cyan, dark grey, and pink respectively.

2. Método de árbol de decisión:

Se encuentra luego de realizar diferentes parametrizaciones, un error bajo del 0.97, en la matriz de confusión solo se tienen dos falsos positivos, lo que da un buen balance.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize Forecast

Classifier

Choose J48 -C 0.7 -M 2 -num-decimal-places 3

Test options

☒ Use training set

☐ Supplied test set

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) medicamento

Start Stop

Result list (right-click for options)

14:10:26 - trees.J48

14:11:42 - trees.J48

14:12:00 - trees.J48

14:12:16 - trees.J48

14:13:23 - trees.J48

14:13:52 - trees.J48

14:14:22 - trees.J48

14:15:27 - trees.J48

14:15:49 - trees.J48

14:16:04 - trees.J48

Classifier output

```

presion_sanguinea = normal
| potasio <= 0.52217
| | sodio <= 0.50943
| | | potasio <= 0.334571: drugY (8.0)
| | | potasio > 0.334571: drugX (12.0)
| | | sodio > 0.50943: drugY (15.0)
| | | potasio > 0.52217: drugX (50.0)

```

Number of Leaves : 23

Size of the tree : 42

Time taken to build model: 0.06 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

	Correctly Classified Instances	Incorrectly Classified Instances	Kappa statistic	Mean absolute error	Root mean squared error	Relative absolute error	Root relative squared error	Total Number of Instances
	448	2	0.9944	0.0029	0.0379	0.8995 %	9.484 %	450

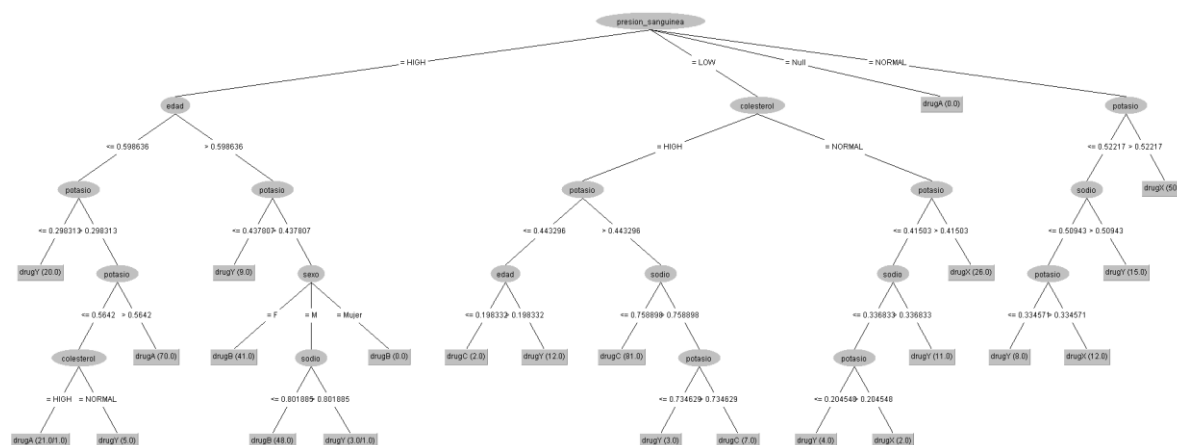
=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.003	0.989	1.000	0.994	0.993	1.000	0.998	drugA
	0.989	0.000	1.000	0.989	0.994	0.993	1.000	1.000	drugB
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	drugC
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	drugX
	0.989	0.003	0.989	0.989	0.989	0.986	1.000	0.998	drugY
Weighted Avg.	0.996	0.001	0.996	0.996	0.996	0.994	1.000	0.999	

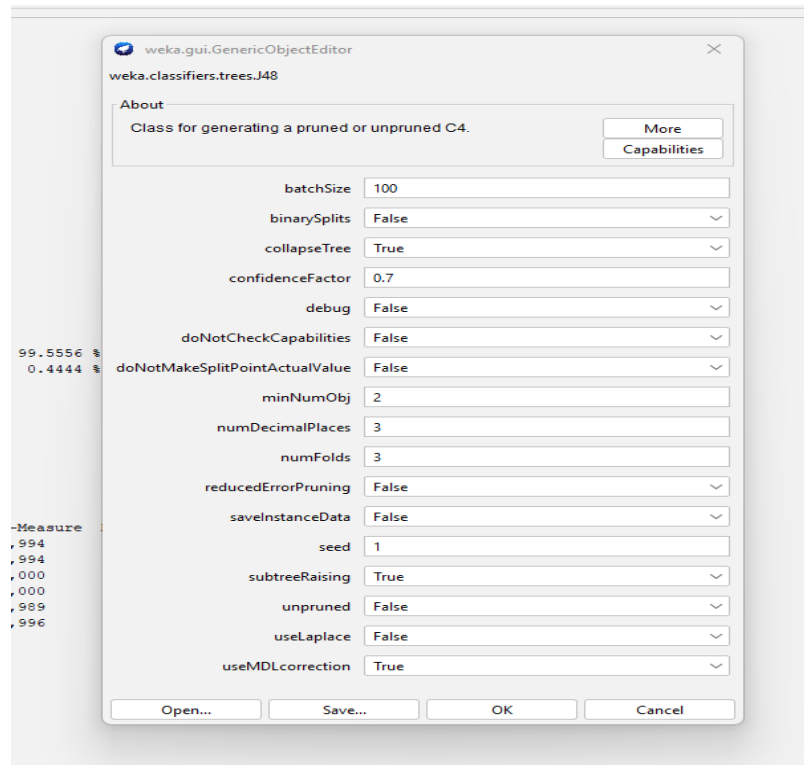
=== Confusion Matrix ===

	a	b	c	d	e	<-- classified as
90	0	0	0	0	0	a = drugA
0	89	0	0	1	0	b = drugB
0	0	90	0	0	0	c = drugC
0	0	0	90	0	0	d = drugX
1	0	0	0	89	0	e = drugY

Gráfico de árbol de decisiones



Configuración de clasificación



Método de redes neuronales:

Se realiza el análisis con la red neuronal y se encuentra un resultado positivo.

Classifier: **MultilayerPerceptron - L03-M02-N500-V0-S0-E20-Hs**

Test options:
☒ Use training set
☐ Supplied test set
☐ Cross-validation
☐ Percentage split
 Folds: 10
 %: 66
 Move options...

(Name) medicamento
 Start Stop

Result list (right-click for options):
 141026 - trees.M0
 141142 - trees.M0
 141200 - trees.M0
 141216 - trees.M0
 141323 - trees.M0
 141332 - trees.M0
 141422 - trees.M0
 141527 - trees.M0
 141549 - trees.M0
 141604 - trees.M0
 142096 - FunctionMultilayerPerceptron

Classifier output:
 Input
 Node 1
 Class drugC
 Input
 Node 2
 Class drugK
 Input
 Node 3
 Class drugF
 Input
 Node 4

Time taken to build model: 0.47 seconds
 Time taken to test model on training data: 0 seconds
 Evaluation on training set ==
 Summary ==
 Correctly Classified Instances 450 100 %
 Incorrectly Classified Instances 0 0 %
 Kappa statistic 1
 Mean Absolute error 0.0038
 Root Mean Squared error 0.01
 Relative Absolute error 1.1649 %
 Root Relative Squared error 2.5092 %
 Total Number of Instances 450

Detailed Accuracy By Class ==

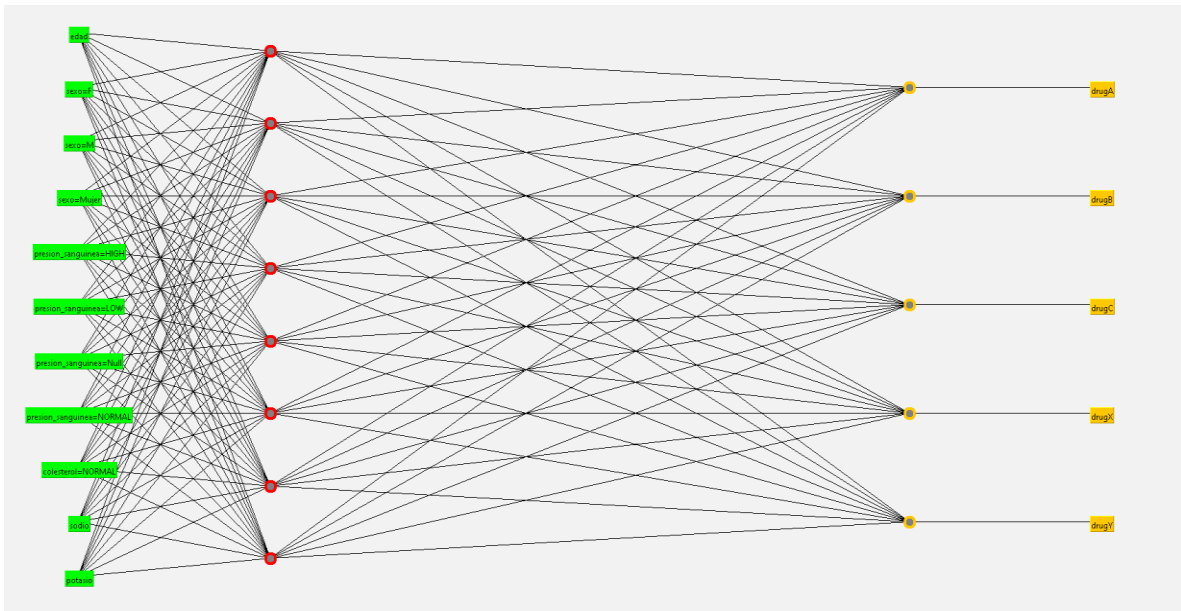
TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	AUC Area	Class
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	drugA
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	drugB
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	drugC
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	drugD
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	drugF
Weighted Avg.	1,000	0,000	1,000	1,000	1,000	1,000	1,000	

Confusion Matrix ==
 a b c d e <-- classified as
 90 0 0 0 0 | a = drugA
 0 90 0 0 0 | b = drugB
 0 0 90 0 0 | c = drugC
 0 0 0 90 0 | d = drugD
 0 0 0 0 90 | e = drugF

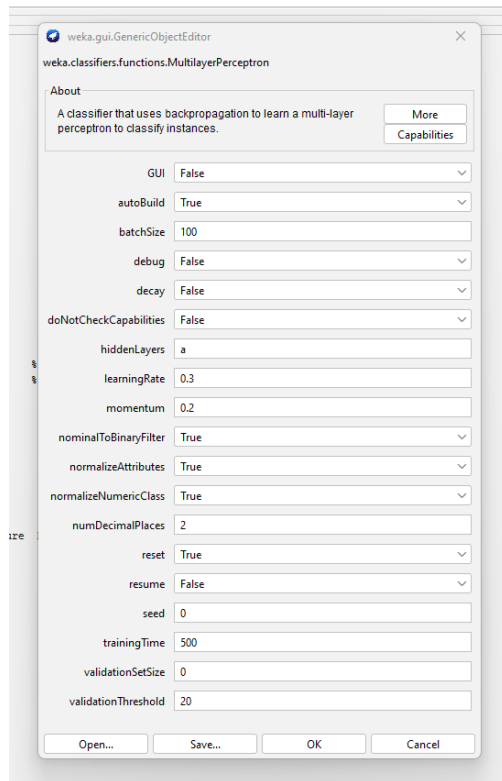
Status: OK

Log

2:22 p.m.
12/04/2025



Configuración de clasificación.



Máquinas de soporte vectorial:

Se encuentra un resultado optimo

Classifier: **SMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -M 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 3.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4"**

Use training set: ☒ Supplied test set: ☐ Cross-validation: ☐ Percentage split: ☐

Number of support vectors: 20
Number of kernel evaluations: 11121 (91.49% cached)

Time taken to build model: 0.05 seconds
Time taken to test model on training data: 0.01 seconds

=== Summary ===

Correctly Classified Instances	490	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0.24		
Root mean squared error	0.3164		
Relative absolute error	76	%	
Root relative squared error	78.8529	%	
Total Number of Instances	490		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	AUC Area	Class
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	drugA
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	drugB
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	drugC
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	drugX
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	drugY
Weighted Avg.	1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	

=== Confusion Matrix ===

	a	b	c	d	e	<-- classified as
90	0	0	0	0	1	a = drugA
0	90	0	0	0	1	b = drugB
0	0	90	0	0	1	c = drugC
0	0	0	90	0	1	d = drugX
0	0	0	0	90	1	e = drugY

Configuración de clasificación

1116 > * X]
1793 > * X]
103 > * X]

weka.gui.GenericObjectEditor

weka.classifiers.functions.SMO

About
Implements John Platt's sequential minimal optimization algorithm for training a support vector classifier.

batchSize: 100
buildCalibrationModels: False
c: 1.0
calibrator: Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**
checksTurnedOff: False
debug: False
doNotCheckCapabilities: False
epsilon: 1.0E-12
filterType: Normalize training data
kernel: Choose **PolyKernel -E 3.0 -C 250007**
numDecimalPlaces: 2
numFolds: -1
randomSeed: 1
toleranceParameter: 0.001

weka.gui.GenericObjectEditor

weka.classifiers.functions.supportVector.PolyKernel

About
The polynomial kernel: $K(x, y) = \langle x, y \rangle^p$ or $K(x, y) = (\langle x, y \rangle + 1)^p$

cacheSize: 250007
debug: False
exponent: **3.0**
useLowerOrder: False

Open... Save... OK Cancel

Regresión

Classifier: **SimpleLogistic -10-M-500-H-50-W-0.0**

Test options:
☒ Use training set
☐ Supplied test set
☐ Cross-validation Folds: 10
☐ Percentage split %: 66
 More options...

Start Stop

Result list (right-click for options):

- 141026 - trees.J48
- 141142 - trees.J48
- 141200 - trees.J48
- 141216 - trees.J48
- 141233 - trees.J48
- 141352 - trees.J48
- 141422 - trees.J48
- 141527 - trees.J48
- 141549 - trees.J48
- 141604 - trees.J48
- 142206 - Functions.MulitayerPerceptron
- 143041 - Functions.SMO
- 143122 - Functions.SMO
- 143134 - Functions.SMO
- 143202 - Functions.SMO
- 143212 - Functions.SMO
- 143220 - Functions.SMO
- 143228 - Functions.SMO
- 143330 - Functions.SMO
- 143350 - Functions.SMO
- 143407 - Functions.SMO
- 143809 - Functions.SimpleLogistic**

Classifier output:

```

[precision_meanquinea=HIGH] * 4.3 +
[colesterol=HIGH] * 4.75 +
[rodicio] * -7.75 +
[potasio] * 10.37

Class drugF :
[sexo=F] * -0.53 +
[colesterol=HIGH] * 1.01 +
[rodicio] * 26.66 +
[potasio] * -52.48
  
```

Time taken to build model: 0.22 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

	Correctly Classified Instances	450	100	%
Incorrectly Classified Instances	0	0	%	
Kappa statistic	1			
Mean absolute error	0.0031			
Root mean squared error	0.0205			
Relative absolute error	0.9463 %			
Root relative squared error	5.1365 %			
Total Number of Instances	450			

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	BWC Area	Class
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	drugA
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	drugB
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	drugC
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	drugK
1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	drugT
Weighted Avg.	1,000	0,000	1,000	1,000	1,000	1,000	1,000	1,000	

=== Confusion Matrix ===

```

a b c d e <-- classified as
90 0 0 0 0 | a = drugA
0 90 0 0 0 | b = drugB
0 0 90 0 0 | c = drugC
0 0 0 90 0 | d = drugK
0 0 0 0 90 | e = drugT
  
```

Status: OK

Clases resultado

Classifier: **SimpleLogistic -10-M-500-H-50-W-0.0**

Test options:
☒ Use training set
☐ Supplied test set
☐ Cross-validation Folds: 10
☐ Percentage split %: 66
 More options...

Start Stop

Result list (right-click for options):

- 41026 - trees.J48
- 41142 - trees.J48
- 41200 - trees.J48
- 41216 - trees.J48
- 41233 - trees.J48
- 41352 - trees.J48
- 41422 - trees.J48
- 41527 - trees.J48
- 41549 - trees.J48
- 41604 - trees.J48
- 42206 - Functions.MulitayerPerceptron
- 43041 - Functions.SMO
- 43122 - Functions.SMO
- 43134 - Functions.SMO
- 43202 - Functions.SMO
- 43212 - Functions.SMO
- 43220 - Functions.SMO
- 43233 - Functions.SMO
- 43330 - Functions.SMO
- 43407 - Functions.SMO
- 43809 - Functions.SimpleLogistic**

Classifier output:

```

=== Classifier model (full training set) ===

SimpleLogistic:

Class drugA :
-2.41 +
[sexo=M] * -17.09 +
[sexo=F] * 0.56 +
[precision_meanquinea=HIGH] * 12.65 +
[colesterol=HIGH] * -5.11 +
[rodicio] * -8.7 +
[potasio] * 9.37

Class drugB :
-26.59 +
[sexo=M] * 23.72 +
[sexo=F] * 1.78 +
[sexo=M] * -0.61 +
[precision_meanquinea=HIGH] * 10.89 +
[colesterol=HIGH] * -5.62 +
[rodicio] * -0 +
[potasio] * 3.08

Class drugC :
-16.95 +
[sexo=M] * -2.59 +
[sexo=F] * 1.35 +
[sexo=M] * -0.6 +
[precision_meanquinea=LOW] * 14.17 +
[colesterol=HIGH] * -13.45 +
[rodicio] * -16.77 +
[potasio] * 25.65

Class drugK :
-3.56 +
[sexo=M] * 2.41 +
[sexo=M] * -0.11 +
[precision_meanquinea=HIGH] * -15.45 +
[precision_meanquinea=HIGH] * 4.3 +
[colesterol=HIGH] * 4.75 +
[rodicio] * -7.75 +
[potasio] * 10.37

Class drugT :
21 +
[sexo=M] * -0.53 +
[colesterol=HIGH] * 1.01 +
[rodicio] * 26.66 +
[potasio] * -52.48
  
```

Configuración de caracterización

weka.gui.GenericObjectEditor

weka.classifiers.functions.SimpleLogistic

About

Classifier for building linear logistic regression models.

More

Capabilities

batchSize 100

debug False

doNotCheckCapabilities False

errorOnProbabilities False

heuristicStop 50

maxBoostingIterations 500

numBoostingIterations 0

numDecimalPlaces 2

useAIC False

useCrossValidation True

weightTrimBeta 0.0

Open... Save... OK Cancel

Método Bayesiano

Classifier: NaiveBayes - num-decimal-places: 10

Test options:

Use training set: ☒ Set...

Supplied test set:

Cross-validation: Folds: 10

Percentage split: % 66

More options...

Model: medicamento

Start Stop

Result list (right-click for options):

- 43122 - Functions.SMO
- 43134 - Functions.SMO
- 43202 - Functions.SMO
- 43212 - Functions.SMO
- 43220 - Functions.SMO
- 43228 - Functions.SMO
- 43330 - Functions.SMO
- 43350 - Functions.SMO
- 43407 - Functions.SMO
- 43609 - Functions.SimpleLogistic
- 44134 - bayes.NaiveBayes
- 44242 - bayes.NaiveBayes
- 44251 - bayes.NaiveBayes
- 44303 - bayes.NaiveBayes
- 44318 - bayes.NaiveBayes
- 44331 - bayes.NaiveBayes
- 44345 - bayes.NaiveBayes
- 44355 - bayes.NaiveBayes
- 44407 - bayes.NaiveBayes
- 44430 - bayes.NaiveBayes
- 44520 - Functions.MultilayerPerceptron
- 44714 - Functions.MultilayerPerceptron
- 44726 - bayes.NaiveBayes
- 44743 - bayes.NaiveBayes
- 44800 - bayes.NaiveBayes
- 44809 - bayes.NaiveBayes
- 44817 - bayes.NaiveBayes
- 44826 - bayes.NaiveBayes
- 44835 - bayes.NaiveBayes
- 44902 - bayes.NaiveBayes
- 44913 - bayes.NaiveBayes
- 44922 - bayes.NaiveBayes
- 44936 - bayes.NaiveBayes
- 44948 - bayes.NaiveBayes
- 44950 - bayes.NaiveBayes
- 44959 - bayes.NaiveBayes

Time taken to build model: 0 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances 429 95.3333 %

Incorrectly Classified Instances 21 4.6667 %

Rapids statistic 0.9417

Mean absolute error 0.0485

Root mean squared error 0.1316

Relative absolute error 15.1638 %

Root relative squared error 32.9024 %

Total Number of Instances 450

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	AUC Area	Class
0 0 0 0 2 1	0.978	0.008	0.967	0.978	0.972	0.945	0.995	0.994	drugA
0 90 0 0 1 1	1.000	0.006	0.978	1.000	0.989	0.994	0.999	0.996	drugB
0 0 0 0 0 1	0.978	0.008	0.967	0.978	0.972	0.945	0.995	0.992	drugC
0 0 0 0 2 1	0.956	0.014	0.945	0.956	0.950	0.930	0.995	0.994	drugD
0 0 0 0 4 1	0.956	0.022	0.906	0.956	0.920	0.922	0.965	0.953	drugE
3 2 2 5 7 1	0.953	0.012	0.953	0.953	0.953	0.941	0.996	0.996	

Weighted Avg.

=== Confusion Matrix ===

a b c d e <-- classified as

0 0 0 0 2 1 a = drugA

0 90 0 0 1 1 b = drugB

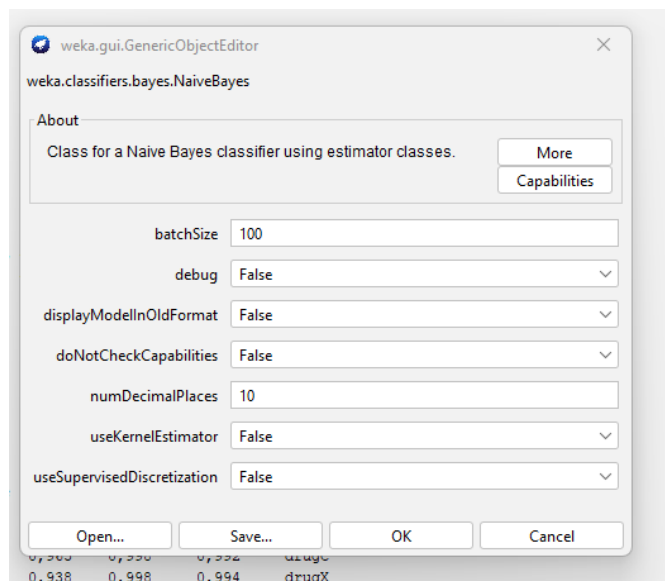
0 0 0 0 2 1 c = drugC

0 0 0 0 4 1 d = drugD

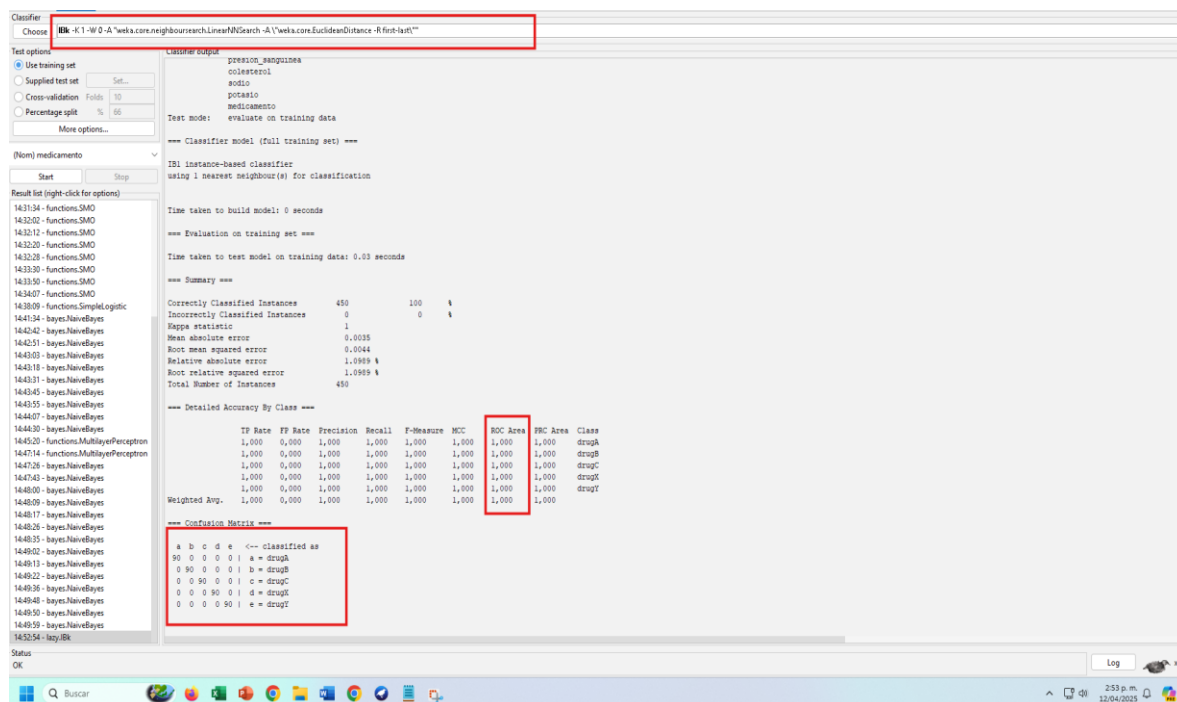
3 2 2 5 7 1 e = drugE

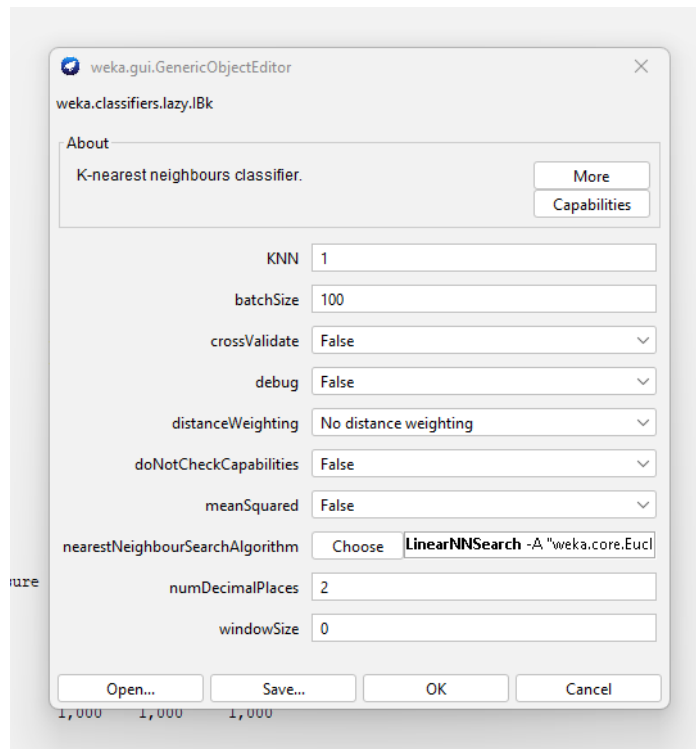
Log

Configuración de caracterización



Método basado en ejemplos





Series de Tiempo

NO aplica porque no tenemos fechas en la estructura

3. Luego de un exhaustivo análisis de los distintos modelos, tomamos la decisión de usar Redes neuronales porque es el método más robusto para solucionar al problema planteado y responde de manera correcta.
4. Luego de realizar la prueba con los nuevos datos de **pacientes_nuevos** encontramos que con el método de redes neuronales y árbol de métodos, el resultado de predicción es igual.

Choose

J48 -C 0.7 -M 2 -num-decimal-places 3

Test option

☐ Use training set

☒ Supplied test set

☐ Cross-validation

☐ Percentage split

Folds

10

%

66

More options...

(Nom) medicamento

Start

Stop

Result list (right-click for options)

14:45:20 - functions.MultilayerPerceptron

14:47:14 - functions.MultilayerPerceptron

14:47:26 - bayes.NaiveBayes

14:47:43 - bayes.NaiveBayes

14:48:00 - bayes.NaiveBayes

14:48:09 - bayes.NaiveBayes

14:48:17 - bayes.NaiveBayes

14:48:26 - bayes.NaiveBayes

14:48:35 - bayes.NaiveBayes

14:49:02 - bayes.NaiveBayes

14:49:13 - bayes.NaiveBayes

14:49:22 - bayes.NaiveBayes

14:49:36 - bayes.NaiveBayes

14:49:48 - bayes.NaiveBayes

14:49:50 - bayes.NaiveBayes

14:49:59 - bayes.NaiveBayes

14:52:54 - lazy.IBk

15:34:19 - misc.InputMappedClassifier

15:40:47 - lazy.IBk

15:41:09 - lazy.IBk

15:41:29 - lazy.IBk

15:41:42 - lazy.IBk

15:42:10 - functions.MultilayerPerceptron

15:42:22 - functions.MultilayerPerceptron

15:42:50 - functions.MultilayerPerceptron

15:43:33 - functions.MultilayerPerceptron

15:46:27 - functions.MultilayerPerceptron

15:46:44 - functions.MultilayerPerceptron

15:46:57 - functions.MultilayerPerceptron

15:48:54 - misc.InputMappedClassifier

15:49:37 - functions.MultilayerPerceptron

15:49:49 - functions.MultilayerPerceptron

15:55:55 - functions.MultilayerPerceptron

16:00:06 - trees.J48

Classifier output

input

Node 4

Time taken to build model: 0.39 seconds

=== Predictions on test set ===

inst#	actual	predicted	error	prediction
1	1:?	5:drugY	0.935	
2	1:?	5:drugY	0.937	
3	1:?	5:drugY	0.933	
4	1:?	5:drugY	0.933	
5	1:?	5:drugY	0.933	
6	1:?	5:drugY	0.977	
7	1:?	5:drugY	0.943	
8	1:?	5:drugY	0.999	
9	1:?	5:drugY	0.939	
10	1:?	5:drugY	0.999	
11	1:?	5:drugY	0.992	

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0.01 seconds

=== Summary ===

Total Number of Instances0

Ignored Class Unknown Instances11

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	?	?	?	?	?	?	?	?	drugA
	?	?	?	?	?	?	?	?	drugB
	?	?	?	?	?	?	?	?	drugC
	?	?	?	?	?	?	?	?	drugX
	?	?	?	?	?	?	?	?	drugY
Weighted Avg.	?	?	?	?	?	?	?	?	

=== Confusion Matrix ===

a b c d e <-- classified as

0 0 0 0 0 | a = drugA

0 0 0 0 0 | b = drugB

0 0 0 0 0 | c = drugC

0 0 0 0 0 | d = drugX

0 0 0 0 0 | e = drugY

Choose

J48 -C 0.7 -M 2 -num-decimal-places 3

Test options

Use training set

Supplied test set

Cross-validation

Percentage split

Set...

Folds

%

10

66

More options...

(Nom) medicamento

Start

Stop

Result list (right-click for options)

14:45:20 - functions.MultilayerPerceptron

14:47:14 - functions.MultilayerPerceptron

14:47:26 - bayes.NaiveBayes

14:47:43 - bayes.NaiveBayes

14:48:00 - bayes.NaiveBayes

14:48:09 - bayes.NaiveBayes

14:48:17 - bayes.NaiveBayes

14:48:26 - bayes.NaiveBayes

14:48:35 - bayes.NaiveBayes

14:49:02 - bayes.NaiveBayes

14:49:13 - bayes.NaiveBayes

14:49:22 - bayes.NaiveBayes

14:49:36 - bayes.NaiveBayes

14:49:48 - bayes.NaiveBayes

14:49:50 - bayes.NaiveBayes

14:49:59 - bayes.NaiveBayes

14:52:54 - lazy.IBk

15:34:19 - misc.InputMappedClassifier

15:40:47 - lazy.IBk

15:41:09 - lazy.IBk

15:41:29 - lazy.IBk

15:41:42 - lazy.IBk

15:42:10 - functions.MultilayerPerceptron

15:42:22 - functions.MultilayerPerceptron

15:42:50 - functions.MultilayerPerceptron

15:43:33 - functions.MultilayerPerceptron

15:46:27 - functions.MultilayerPerceptron

15:46:44 - functions.MultilayerPerceptron

15:46:57 - functions.MultilayerPerceptron

15:48:54 - misc.InputMappedClassifier

15:49:37 - functions.MultilayerPerceptron

15:49:49 - functions.MultilayerPerceptron

15:55:55 - functions.MultilayerPerceptron

16:00:06 - trees.J48

16:00:29 - trees.J48

16:00:33 - trees.J48

Classifier output

Size of the tree : 42

Time taken to build model: 0.06 seconds

=== Predictions on test set ===

inst#	actual	predicted	error	prediction
1	1:?	5:drugY	1	
2	1:?	5:drugY	1	
3	1:?	5:drugY	1	
4	1:?	5:drugY	1	
5	1:?	5:drugY	1	
6	1:?	5:drugY	1	
7	1:?	5:drugY	1	
8	1:?	5:drugY	1	
9	1:?	5:drugY	1	
10	1:?	5:drugY	1	
11	1:?	5:drugY	1	

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0 seconds

=== Summary ===

Total Number of Instances 0

Ignored Class Unknown Instances 11

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	?	?	?	?	?	?	?	?	drugA
	?	?	?	?	?	?	?	?	drugB
	?	?	?	?	?	?	?	?	drugC
	?	?	?	?	?	?	?	?	drugX
	?	?	?	?	?	?	?	?	drugY
Weighted Avg.	?	?	?	?	?	?	?	?	

=== Confusion Matrix ===

a b c d e <-- classified as

0 0 0 0 0 | a = drugA

0 0 0 0 0 | b = drugB

0 0 0 0 0 | c = drugC

0 0 0 0 0 | d = drugX

0 0 0 0 0 | e = drugY